

Example: Suppose that $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

Since these are the real and imaginary components, respectively, of the entire Function $f(z) = z^2$, then v is a harmonic conjugate of u throughout the plane. But u cannot be a harmonic conjugate of v then function $2xy + i(x^2 - y^2)$ is not analytic anywhere.

METHOD TO FIND HARMONIC CONJUGATE:

For a given real valued function $u(x, y)$

- Find u_x and then using CR equation write $u_x = v_y$
- Integrate w.r.to 'y' to get $v(x, y)$ but a constant also appear.
- Differentiate w.r.to 'x' to get v_x
- To find constant appear in previous step using CR equation write $v_x = -u_y$
- After putting the value of constant finally get $v(x, y)$

Example: Find harmonic conjugate for $U(x, y) = e^x \cos y$

Solution: Given that $U(x, y) = e^x \cos y$

$$\Rightarrow U_x = e^x \cos y \Rightarrow U_x = e^x \cos y = V_y$$

Integrate w.r.to 'y' $\Rightarrow V(x, y) = e^x \sin y + g(x)$

Differentiate w.r.to 'x' $\Rightarrow V_x(x, y) = e^x \sin y + g'(x)$

using CR equation $\Rightarrow V_x = -U_y \Rightarrow -U_y = e^x \sin y + g'(x)$

$$\Rightarrow V_x = -U_y \Rightarrow -e^x \sin y = e^x \sin y + g'(x) \Rightarrow g'(x) = 0 \Rightarrow g(x) = c$$

After putting the value of constant finally get

$$\Rightarrow V(x, y) = e^x \sin y + c$$

Example: Prove that the function $U(x, y) = e^x \cos y$ is harmonic. Find its Corresponding conjugate and construct the original function.

Solution: Given that $U(x, y) = e^x \cos y$

This function is already proved in previous steps that this is Harmonic

Also we find its corresponding conjugates.

To construct original function.

Since $\Rightarrow f(z) = U(x, y) + iV(x, y)$

$$\Rightarrow f(z) = e^x \cos y + ie^x \sin y + ic \Rightarrow f(z) = e^x (\cos y + i \sin y) + c'$$

$$\Rightarrow f(z) = e^x e^{iy} + c' = e^{x+iy} + c' = e^z + c' \Rightarrow f(z) = e^z + c'$$

Exercise 23: Prove that the given function is harmonic. Find its Corresponding conjugate and construct the original function.

- $u(x, y) = x^3 - 3xy^2$
- $u(x, y) = e^x(x \cos y - y \sin y)$
- $u(r, \theta) = r^n \cos n\theta$
- $u(x, y) = 2x(1 - y)$
- $u(x, y) = 2x - x^3 + 3xy^2$
- $u(x, y) = \sinh x \sin y$
- $u(x, y) = y/(x^2 + y^2)$

Example: Prove that a harmonic function satisfies the formal differential

$$\text{Equation } \frac{\partial^2 U}{\partial z \partial \bar{z}} = 0$$

Solution: Since U is harmonic therefor $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = 0 \dots\dots\dots(i)$

$$\text{Since } x = \frac{z+\bar{z}}{2} \text{ and } y = \frac{z-\bar{z}}{2i}$$

$$\begin{aligned} \text{Therefore } \frac{\partial U}{\partial \bar{z}} &= \frac{\partial U}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial U}{\partial y} \frac{\partial y}{\partial \bar{z}} \Rightarrow \frac{\partial U}{\partial \bar{z}} = \frac{1}{2} \frac{\partial U}{\partial x} + \frac{1}{2i} \frac{\partial U}{\partial y} \\ \Rightarrow \frac{\partial^2 U}{\partial z \partial \bar{z}} &= \frac{\partial}{\partial z} \left(\frac{\partial U}{\partial \bar{z}} \right) = \frac{1}{2} \left(\frac{\partial^2 U}{\partial x^2} \frac{\partial x}{\partial \bar{z}} + \frac{\partial^2 U}{\partial y \partial x} \frac{\partial y}{\partial \bar{z}} \right) - \frac{1}{2i} \left(\frac{\partial^2 U}{\partial y \partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial^2 U}{\partial y^2} \frac{\partial y}{\partial \bar{z}} \right) \\ \Rightarrow \frac{\partial^2 U}{\partial z \partial \bar{z}} &= \frac{1}{2} \left(\frac{1}{2} \frac{\partial^2 U}{\partial x^2} - \frac{i}{2} \frac{\partial^2 U}{\partial y \partial x} \right) - \frac{1}{2i} \left(\frac{1}{2} \frac{\partial^2 U}{\partial y \partial x} + \frac{1}{2i} \frac{\partial^2 U}{\partial y^2} \right) \\ \Rightarrow \frac{\partial^2 U}{\partial z \partial \bar{z}} &= \frac{1}{4} \frac{\partial^2 U}{\partial x^2} - \frac{i}{4} \frac{\partial^2 U}{\partial y \partial x} + \frac{i}{4} \frac{\partial^2 U}{\partial y \partial x} + \frac{1}{4} \frac{\partial^2 U}{\partial y^2} \\ \Rightarrow \frac{\partial^2 U}{\partial z \partial \bar{z}} &= \frac{1}{4} \left(\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} \right) \Rightarrow \frac{\partial^2 U}{\partial z \partial \bar{z}} = 0 \quad \therefore \text{ using } (i) \end{aligned}$$

METHOD OF CONSTRUCTING AN ANALYTIC FUNCTION;

For a given real valued function $u(x, y)$

- i. Find $u_x(z, 0)$ and $u_y(z, 0)$
- ii. $f'(z) = u_x(z, 0) - iu_y(z, 0)$
- iii. To find $f(z)$ integrate $f'(z)$

Example: Verify that the function $U(x, y) = \text{Sin}x \text{Cos}y$ is harmonic. And find analytic function.

Solution: Given that $U(x, y) = \text{Sin}x \text{Cos}y$
 $\Rightarrow U_x = \text{Cos}x \text{Cos}y \Rightarrow U_{xx} = -\text{Sin}x \text{Cos}y$
 also $\Rightarrow U_y = \text{Sin}x \text{Sin}y \Rightarrow U_{yy} = \text{Sin}x \text{Cos}y$
 $\Rightarrow U_{xx} + U_{yy} = -\text{Sin}x \text{Cos}y + \text{Sin}x \text{Cos}y = 0 \Rightarrow U_{xx} + U_{yy} = 0$
 $\Rightarrow U(x, y) = \text{Sin}x \text{Cos}y$ is harmonic.

Now $u_x(z, 0) = \text{Cos}z$ and $u_y(z, 0) = 0$

Then $f'(z) = u_x(z, 0) - iu_y(z, 0) = \text{Cos}z - i0 \Rightarrow f'(z) = \text{Cos}z$

Integrate w.r.to 'z' $\Rightarrow f(z) = \text{Sin}z$ which is required.

Exercise 24: Verify that the given function is harmonic.

And find analytic function.

- i. $u(x, y) = \frac{x}{x^2+y^2}$
- ii. $v(x, y) = x^3 - 3xy^2 - 5y$
- iii. $u(x, y) = e^{-x}[(x^2 - y^2)\text{Cos}y + 2xy\text{Sin}y]$
- iv. $u(x, y) = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$
- v. $u(x, y) = \log \sqrt{x^2 + y^2}$ in the disk $|z - 1| < 1$
- vi. $u(x, y) = \frac{\text{Sin}2x}{\text{Cosh}2y + \text{Cos}2x}$
- vii. $u(x, y) = x^3 - 3xy^2 + 3x + 1$
- viii. $u(x, y) = y^3 - 3x^2y$

LEVEL CURVES: A level curve of a real valued function $U(x,y)$ defined in a Domain D is given by the locus of a point (x,y) in D such that $U(x,y) = C$ Where C is constant.

Likewise we can consider a level curve of $V(x,y)$ in D

i.e. $V(x,y) = K$ Where K is constant.

Example: Sketch the level curves of the function $f(z) = z^2$; $z \neq 0$

Solution: Given $f(z) = z^2 = x^2 - y^2 + i2xy$

Put $U(x,y) = x^2 - y^2 = C_1$ and $V(x,y) = 2xy = C_2$

These are the required level curves and are Rectangular Hyperbolas.

ORTHOGONAL SYSTEM: The families of curves $U(x,y) = C_1$ and $V(x,y) = C_2$ Are said to form an orthogonal system if the curves intersect at right angles at each Of their point of intersection.

CONDITION TO FIND ORTHOGONALITY OF LEVEL CURVES

Consider $U(x,y) = C_1$

Differentiating we get $\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\partial U}{\partial x} / \frac{\partial U}{\partial y} \Rightarrow m_1 = -\frac{\partial U}{\partial x} / \frac{\partial U}{\partial y}$

Also $V(x,y) = C_2$

Differentiating we get $\frac{\partial V}{\partial x} + \frac{\partial V}{\partial y} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\partial V}{\partial x} / \frac{\partial V}{\partial y} \Rightarrow m_2 = -\frac{\partial V}{\partial x} / \frac{\partial V}{\partial y}$

Now the two families of curves intersect orthogonally if

$$m_1 m_2 = -1 \Rightarrow \left(-\frac{\partial U}{\partial x} / \frac{\partial U}{\partial y}\right) \left(-\frac{\partial V}{\partial x} / \frac{\partial V}{\partial y}\right) = -1$$

$$\frac{\partial U}{\partial x} \frac{\partial V}{\partial x} + \frac{\partial U}{\partial y} \frac{\partial V}{\partial y} = 0 \text{ this is required condition.}$$

Example: If $w = \frac{1}{z}$ then show that level curves $U = C_1$ and $V = C_2$

Are orthogonal circles which pass through the origin and have their centers On x -axis and y -axis.

Solution: Given $w = \frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} + i \frac{-y}{x^2+y^2}$

Here $U(x,y) = \frac{x}{x^2+y^2}$ and $V(x,y) = \frac{-y}{x^2+y^2}$

When $U = \frac{x}{x^2+y^2} = C_1 \Rightarrow C_1 = \frac{x}{x^2+y^2} \Rightarrow C_1(x^2 + y^2) - x = 0$ and pass through

Origin having centre $\left(\frac{1}{2}C_1, 0\right)$

also $V = \frac{-y}{x^2+y^2} = C_2 \Rightarrow C_2 = \frac{-y}{x^2+y^2} \Rightarrow C_2(x^2 + y^2) + y = 0$ and pass through

Origin having centre $\left(0, \frac{1}{2}C_2\right)$

For Orthogonality:

$$\frac{\partial U}{\partial x} \frac{\partial V}{\partial x} + \frac{\partial U}{\partial y} \frac{\partial V}{\partial y} = \frac{y^2-x^2}{(x^2+y^2)^2} \frac{2xy}{x^2+y^2} + \frac{-2xy}{x^2+y^2} \frac{y^2-x^2}{(x^2+y^2)^2} = 0$$

Thus level curves are orthogonal.

Example: If $f(z) = \frac{z+4}{z-4}$ then find the level curves $U = C_1$ and $V = C_2$
Also verify that level curves form an orthogonal system.

Solution: Given $f(z) = \frac{z+4}{z-4} = \frac{(x+4)+iy}{(x-4)+iy} = \frac{(x^2+y^2-16)+i(-8y)}{(x-4)^2+y^2}$

Here $U(x, y) = \frac{x^2+y^2-16}{(x-4)^2+y^2}$ and $V(x, y) = \frac{-8y}{(x-4)^2+y^2}$

When $U = \frac{x^2+y^2-16}{(x-4)^2+y^2} = C_1 \Rightarrow C_1(x^2 + y^2 - 8x + 16) - x^2 - y^2 + 16 = 0$
 $\Rightarrow (C_1 - 1)x^2 + (C_1 - 1)y^2 - 8C_1x + 16C_1 + 16 = 0$ which is circle.

Origin having centre $\left(\frac{1}{2}C_1, 0\right)$

Also $V = \frac{-8y}{(x-4)^2+y^2} = C_2 \Rightarrow C_2(x^2 + y^2 - 8x + 16) + 8y = 0$ which is also circle.

For Orthogonality:

$$\frac{\partial U}{\partial x} \frac{\partial V}{\partial x} + \frac{\partial U}{\partial y} \frac{\partial V}{\partial y} = \frac{-8x^2+8y^2+64x-128}{[(x-4)^2+y^2]^2} \cdot \frac{16(x-4)}{[(x-4)^2+y^2]^2} + \frac{-16(x-4)}{[(x-4)^2+y^2]^2} \cdot \frac{8x^2+8y^2+64x-128}{[(x-4)^2+y^2]^2} = 0$$

Thus level curves are orthogonal.

Example: If $f(z) = U + iV$ be an analytic function of $z = x+iy$ then show that
The curves $U = C_1$ and $V = C_2$ intersect at right angle.

Solution: Given $f(z)$ to be analytic. Then CR equations hold.

i.e. $\frac{\partial U}{\partial x} = \frac{\partial V}{\partial y} \dots \dots (i)$ and $\frac{\partial V}{\partial x} = -\frac{\partial U}{\partial y} \dots \dots (ii)$

multiplying (i) and (ii) we get $\frac{\partial U}{\partial x} \frac{\partial V}{\partial x} = -\frac{\partial U}{\partial y} \frac{\partial V}{\partial y}$

$\frac{\partial U}{\partial x} \frac{\partial V}{\partial x} + \frac{\partial U}{\partial y} \frac{\partial V}{\partial y} = 0$ required condition of orthogonality of two curves.

Problem: Prove that an analytic function with constant modulus is constant.

Solution: Since f is analytic in D , then it must satisfy the CR equations

i.e. $u_x = v_y$, and $v_x = -u_y$ and also $|f(z)| = C' \Rightarrow u^2 + v^2 = C \dots \dots (i)$

differentiating (i) w.r.to $x \Rightarrow 2uu_x + 2vv_x = 0 \Rightarrow uu_x + vv_x = 0 \dots \dots (ii)$

likewise differentiating (i) w.r.to $y \Rightarrow 2uu_y + 2vv_y = 0 \Rightarrow uu_y + vv_y = 0$

$\Rightarrow -uv_x + vu_x = 0 \dots \dots (iii)$ using CR equations.

Multiplying (ii) by 'u' and (iii) by 'v' and then adding we get

$\Rightarrow u_x(u^2 + v^2) = 0 \Rightarrow (u^2 + v^2) \neq 0$ using (i) then $u_x = 0$

$\Rightarrow u_x = 0 = v_y \Rightarrow$ 'u' is independent of x and 'v' is independent of y .

Multiplying (ii) by 'v' and (iii) by 'u' and then subtracting we get

$\Rightarrow v_x(u^2 + v^2) = 0 \Rightarrow (u^2 + v^2) \neq 0$ using (i) then $v_x = 0$

$\Rightarrow v_x = 0 = -u_y \Rightarrow$ 'v' is independent of x and 'u' is independent of y .

Thus 'u' and 'v' are both independent of x and y .

Therefore u and v are constants.

Ultimately $f(z) = u + iv = \text{Constant}$

Problem: Investigate the value of z for which $W = U + iV$ is not analytic

When $z = \text{Sinh}U\cos V + i\text{Cosh}U\sin V$

Solution: Given $z = \text{Sinh}U\cos V + i\text{Cosh}U\sin V$

$$\Rightarrow \frac{dz}{dw} = \frac{\partial z}{\partial u} = \text{Cosh}U\cos V + i\text{Sinh}U\sin V \dots \dots \dots (i)$$

$$\text{Also } z^2 = (\text{Sinh}U\cos V + i\text{Cosh}U\sin V)^2$$

$$z^2 = \text{Sin}^2hU\cos^2V - \text{Cos}^2hU\sin^2V + 2i\text{sinh}U\cos V\cosh U\sin V \dots \dots \dots (ii)$$

$$\therefore \text{Cos}^2hU - \text{Sin}^2hU = 1$$

$$(ii) \Rightarrow z^2 = (\text{Cos}^2hU - 1)\cos^2V - (1 + \text{Sin}^2hU)\sin^2V + 2i\text{sinh}U\cos V\cosh U\sin V$$

$$\Rightarrow z^2 = \text{Cos}^2hU\cos^2V - \cos^2V - \sin^2hU - \text{Sin}^2hUV\sin^2V + 2i\text{sinh}U\cos V\cosh U\sin V$$

$$\Rightarrow z^2 = \text{Cos}^2hU\cos^2V - \text{Sin}^2hUV\sin^2V + 2i\text{sinh}U\cos V\cosh U\sin V - (\cos^2V + \sin^2hU)$$

$$\Rightarrow z^2 = (\cosh U\cos V + i\sinh U\sin V)^2 - 1 \Rightarrow z^2 = \left(\frac{dz}{dw}\right)^2 - 1 \Rightarrow \left(\frac{dz}{dw}\right)^2 = 1 + z^2$$

$$\Rightarrow \frac{dw}{dz} = \pm \sqrt{1 + z^2} \Rightarrow \frac{dz}{dw} = \pm \frac{1}{\sqrt{1 + z^2}}$$

$$\frac{dw}{dz} \text{ does not exist when } 1 + z^2 = 0 \Rightarrow z^2 = -1 \Rightarrow z = \pm i$$

Therefore function is not analytic when $z = \pm i$ where $z = \text{Sinh}U\cos V + i\text{Cosh}U\sin V$

Exercise 25: (visit @ Youtube “learning with Usman Hamid”)

Investigate the value of z for which $W = f(z)$ ceases to be analytic where

i. $z = e^{-v}(-\cos u + i\sin u)$

ii. $z = \text{Sin}u\text{Cosh}v + i\text{Cos}u\text{sinh}v$

UNIQUELY DETERMINED ANALYTIC FUNCTIONS

Suppose that a function f is analytic throughout a domain D and $f(z) = 0$ at each point z of a domain or line segment contained in D .

Then $f(z) \equiv 0$ in D ; that is, $f(z)$ is identically equal to zero throughout D .

Remark:

- A function that is analytic in a domain D is uniquely determined over D by its values in a domain, or along a line segment, contained in D .
- **Reflection Principle :** Suppose that a function f is analytic in some domain D which contains a segment of the x axis and whose lower half is the reflection of the upper half with respect to that axis
Then $\overline{f(z)} = f(\bar{z})$

Problem: If $f(z) = U + iV$ is an analytic function of 'z' in any domain then prove that $\nabla^2 |f(z)|^p = p^2 |f(z)|^{p-2} |f'(z)|^2$ also deduce for $p = 2$.

Solution: Suppose $f(z) = U + iV$

$$\begin{aligned} \Rightarrow |f(z)| &= \sqrt{U^2 + V^2} \Rightarrow |f(z)|^p = (U^2 + V^2)^{p/2} \dots\dots\dots(i) \\ \Rightarrow \frac{\partial}{\partial x} |f(z)|^p &= \frac{p}{2} (U^2 + V^2)^{\frac{p}{2}-1} (2UU_x + 2VV_x) = p (U^2 + V^2)^{\frac{p-2}{2}} (UU_x + VV_x) \\ \Rightarrow \frac{\partial}{\partial x} |f(z)|^p &= p (U^2 + V^2)^{\frac{p-2}{2}} (UU_x + VV_x) \\ \Rightarrow \frac{\partial^2}{\partial x^2} |f(z)|^p &= \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} |f(z)|^p \right) = p \left(\frac{p-2}{2} \right) (U^2 + V^2)^{\frac{p-2}{2}-1} (2UU_x + 2VV_x)(UU_x + VV_x) \\ &\quad + p (U^2 + V^2)^{\frac{p-2}{2}} (UU_{xx} + U_x U_x + VV_{xx} + V_x V_x) \\ \Rightarrow \frac{\partial^2}{\partial x^2} |f(z)|^p &= p(p-2) (U^2 + V^2)^{\frac{p-4}{2}} (UU_x + VV_x)^2 \\ &\quad + p (U^2 + V^2)^{\frac{p-2}{2}} (UU_{xx} + U_x^2 + VV_{xx} + V_x^2) \dots\dots\dots(ii) \end{aligned}$$

Similarly

$$\begin{aligned} \Rightarrow \frac{\partial^2}{\partial y^2} |f(z)|^p &= p(p-2) (U^2 + V^2)^{\frac{p-4}{2}} (UU_y + VV_y)^2 \\ &\quad + p (U^2 + V^2)^{\frac{p-2}{2}} (UU_{yy} + U_y^2 + VV_{yy} + V_y^2) \dots\dots\dots(iii) \end{aligned}$$

Now we know that

$$\nabla^2 |f(z)|^p = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^p = \frac{\partial^2}{\partial x^2} |f(z)|^p + \frac{\partial^2}{\partial y^2} |f(z)|^p$$

Then using equation (ii) and (iii) we get

$$\begin{aligned} \nabla^2 |f(z)|^p &= p(p-2) (U^2 + V^2)^{\frac{p-4}{2}} (UU_x + VV_x)^2 \\ &\quad + p (U^2 + V^2)^{\frac{p-2}{2}} (UU_{xx} + U_x^2 + VV_{xx} + V_x^2) \\ &\quad + p(p-2) (U^2 + V^2)^{\frac{p-4}{2}} (UU_y + VV_y)^2 \\ &\quad + p (U^2 + V^2)^{\frac{p-2}{2}} (UU_{yy} + U_y^2 + VV_{yy} + V_y^2) \\ \nabla^2 |f(z)|^p &= p(p-2) (U^2 + V^2)^{\frac{p-4}{2}} [(UU_x + VV_x)^2 + (UU_y + VV_y)^2] \\ &\quad + p (U^2 + V^2)^{\frac{p-2}{2}} (UU_{xx} + U_x^2 + VV_{xx} + V_x^2 + UU_{yy} + U_y^2 + VV_{yy} + V_y^2) \\ \nabla^2 |f(z)|^p &= p(p-2) (U^2 + V^2)^{\frac{p-4}{2}} [(UU_x + VV_x)^2 + (UU_y + VV_y)^2] \\ &\quad + p (U^2 + V^2)^{\frac{p-2}{2}} [U(U_{xx} + U_{yy}) + V(V_{xx} + V_{yy}) + U_x^2 + V_x^2 + U_y^2 + V_y^2] \end{aligned}$$

Since $U_{xx} + U_{yy} = 0 = V_{xx} + V_{yy}$ and Function is analytic. Therefore CR equation Will be satisfied i.e. $U_x = V_y$ and $V_x = -U_y$ or $U_y = -V_x$ then

$$\begin{aligned} \nabla^2 |f(z)|^p &= p(p-2) (U^2 + V^2)^{\frac{p-4}{2}} [(UU_x + VV_x)^2 + (-UV_x + VU_x)^2] \\ &\quad + p (U^2 + V^2)^{\frac{p-2}{2}} [U(0) + V(0) + U_x^2 + V_x^2 + V_x^2 + U_x^2] \end{aligned}$$

$$= p(p-2)(U^2 + V^2)^{\frac{p-4}{2}} \left[U^2 U_x^2 + V^2 V_x^2 + 2UVU_x V_x + U^2 V_x^2 + V^2 U_x^2 - 2UVU_x V_x \right] \\ + p(U^2 + V^2)^{\frac{p-2}{2}} [2U_x^2 + 2V_x^2]$$

$$\nabla^2 |f(z)|^p = p(p-2)(U^2 + V^2)^{\frac{p-4}{2}} [U^2(U_x^2 + V_x^2) + V^2(U_x^2 + V_x^2)] \\ + 2p(U^2 + V^2)^{\frac{p-2}{2}} [U_x^2 + V_x^2]$$

$$\nabla^2 |f(z)|^p = p(p-2)(U^2 + V^2)^{\frac{p-4}{2}} [(U_x^2 + V_x^2)(U^2 + V^2)] \\ + 2p(U^2 + V^2)^{\frac{p-2}{2}} [U_x^2 + V_x^2]$$

$$\nabla^2 |f(z)|^p = p(p-2)(U^2 + V^2)^{\frac{p-2}{2}} (U_x^2 + V_x^2) \\ + 2p(U^2 + V^2)^{\frac{p-2}{2}} [U_x^2 + V_x^2]$$

$$\nabla^2 |f(z)|^p = p(U^2 + V^2)^{\frac{p-2}{2}} (U_x^2 + V_x^2) [p-2+2]$$

$$\nabla^2 |f(z)|^p = p^2 (U^2 + V^2)^{\frac{p-2}{2}} (U_x^2 + V_x^2)$$

$$\nabla^2 |f(z)|^p = p^2 |f(z)|^{p-2} |f'(z)|^2$$

When $p = 2$

$$\nabla^2 |f(z)|^2 = 2^2 |f(z)|^{2-2} |f'(z)|^2$$

$$\nabla^2 |f(z)|^2 = 4 |f'(z)|^2$$

Practice:

If $f(z)$ is an analytic function of 'z' in any domain then prove that

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |Re(f(z))|^2 = 2 |f'(z)|^2$$

Solution: Suppose $f(z) = u + iv$

$$\Rightarrow |Re(f(z))| = u \Rightarrow |Re(f(z))|^2 = u^2 \Rightarrow S = u^2 \text{ (say)}$$

$$\Rightarrow S_x = 2uu_x \Rightarrow S_{xx} = 2(uu_{xx} + u_x^2) \text{ and } S_y = 2uu_y \Rightarrow S_{yy} = 2(uu_{yy} + u_y^2)$$

$$\Rightarrow S_{xx} + S_{yy} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) S = 2(uu_{xx} + u_x^2) + 2(uu_{yy} + u_y^2)$$

$$\Rightarrow \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |Re(f(z))|^2 = 2[u(u_{xx} + u_{yy}) + (u_x^2 + u_y^2)]$$

$$\Rightarrow \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |Re(f(z))|^2 = 2(u_x^2 + u_y^2) \quad \therefore u_{xx} + u_{yy} = 0$$

$$\text{Hence } \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |Re(f(z))|^2 = 2 |f'(z)|^2$$

CHAPTER

3

ELEMENTARY FUNCTIONS

THE EXPONENTIAL FUNCTION

We define here the exponential function e^z by writing

$$e^z = e^x e^{iy} \quad \text{where } (z = x + iy),$$

Where Euler's formula gives $e^{iy} = \cos y + i \sin y$ and y is to be taken in radians.

Remark:

- i. $e^z = \rho e^{i\varphi}$ where $\rho = e^x$ and $\varphi = y$
- ii. $|e^z| = e^x$
- iii. $\arg(e^z) = y + 2n\pi$; $n = 0, \pm 1, \pm 2, \dots$
- iv. $e^{z_1} e^{z_2} = e^{z_1 + z_2}$ (addition theorem)
- v. $\frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}$
- vi. $e^z \neq 0$ as e^x is never zero.
- vii. $e^0 = 1$
- viii. $e^{-z} = \frac{1}{e^z}$
- ix. $\frac{d}{dz} e^z = e^z$ everywhere in the 'z' plane.
- x. $e^{z+2\pi i} = e^z$
- xi. $e^{\pi i} = -1$ and $e^{2\pi i} = 1$
- xii. $e^{(2n+1)\pi i} = -1$; $n = 0, \pm 1, \pm 2, \dots$
- xiii. e^z converges absolutely.

Example: Find numbers $z = x + iy$ such that $e^z = 1 + i$

Solution: Given that $e^z = 1 + i$

$$\Rightarrow e^x e^{iy} = 1 + i$$

$$\text{Then } |e^z| = e^x = \sqrt{2} \Rightarrow x = \ln \sqrt{2} = \frac{1}{2} \ln 2$$

$$\text{and } \arg(e^z) = \theta + 2n\pi = \frac{\pi}{4} + 2n\pi = \left(2n + \frac{1}{4}\right)\pi ; n = 0, \pm 1, \pm 2, \dots$$

now we know that $z = x + iy$

then after putting the values

$$z = x + iy = \frac{1}{2} \ln 2 + \left(2n + \frac{1}{4}\right)\pi i ; n = 0, \pm 1, \pm 2, \dots$$

Property: Prove that $e^z = 1$ iff $z = 2k\pi i$; $k = 0, \pm 1, \pm 2, \dots$

Solution: Consider $e^z = 1$ Then we have to prove $z = 2k\pi i$

Now as $e^z = 1 \Rightarrow e^x e^{iy} = 1 \Rightarrow e^x (\cos y + i \sin y) = 1$

$\Rightarrow e^x \cos y + i e^x \sin y = 1 + 0i$

$e^x \cos y = 1 \dots (i) \quad e^x \sin y = 0 \dots (ii)$

$(ii) \Rightarrow e^x \neq 0 \Rightarrow \sin y = 0 \Rightarrow y = \sin^{-1}(0) \Rightarrow y = n\pi ; n = 0, \pm 1, \pm 2, \dots$

$(i) \Rightarrow e^x \cos y = 1 \Rightarrow e^x \cos n\pi = 1 \Rightarrow e^x (-1)^n = e^0$

$\Rightarrow (-1)^n = 1$ if $n = 2k$; $k = 0, \pm 1, \pm 2, \pm 3, \dots$

Thus $e^x = e^0 \Rightarrow x = 0$ and therefore $y = n\pi = 2k\pi$

Hence $z = x + iy = 2k\pi i$

For Sufficient Condition:

Given $z = 2k\pi i$ then we have to show that $e^z = 1$

Let $e^z = e^{2k\pi i} = \cos 2k\pi + i \sin 2k\pi = 1 + 0$

$\Rightarrow e^z = 1$ as required

Property: Prove that $e^{z_1} = e^{z_2}$ if $z_1 = z_2 + 2k\pi i$

Solution: Consider $e^{z_1} = e^{z_2}$ Then we have to prove $z = 2k\pi i$

Now as $e^{z_1} = e^{z_2} \Rightarrow e^{z_1 - z_2} = 1 \Rightarrow e^{(x_1 - x_2) + i(y_1 - y_2)} = 1$

$\Rightarrow e^{x_1 - x_2} [\cos(y_1 - y_2) + i \sin(y_1 - y_2)] = 1 + 0i$

$e^{x_1 - x_2} \cos(y_1 - y_2) = 1 \dots (i) \quad e^{x_1 - x_2} \sin(y_1 - y_2) = 0 \dots (ii)$

$(ii) \Rightarrow e^{x_1 - x_2} \neq 0 \Rightarrow \sin(y_1 - y_2) = 0 \Rightarrow (y_1 - y_2) = \sin^{-1}(0)$

$\Rightarrow (y_1 - y_2) = n\pi ; n = 0, \pm 1, \pm 2, \dots$

$(i) \Rightarrow e^{x_1 - x_2} \cos(y_1 - y_2) = 1 \Rightarrow e^{x_1 - x_2} \cos n\pi = 1 \Rightarrow e^{x_1 - x_2} (-1)^n = e^0$

$\Rightarrow (-1)^n = 1$ if $n = 2k$; $k = 0, \pm 1, \pm 2, \pm 3, \dots$

Thus $e^{x_1 - x_2} = e^0 \Rightarrow x_1 - x_2 = 0 \Rightarrow x_1 = x_2$

and therefore $z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2) = 0 + i2k\pi$

Hence $z_1 - z_2 = 2k\pi i \Rightarrow z_1 = z_2 + 2k\pi i$

Property: Prove that $e^z = -1$ iff $z = (2k + 1)\pi i$; $k = 0, \pm 1, \pm 2, \dots$

Solution: Consider $e^z = -1$ Then we have to prove $z = (2k + 1)\pi i$

Now as $e^z = -1 \Rightarrow e^x e^{iy} = -1 \Rightarrow e^x (\cos y + i \sin y) = -1$

$\Rightarrow e^x \cos y + i e^x \sin y = -1 + 0i$

$e^x \cos y = -1 \dots (i) \quad e^x \sin y = 0 \dots (ii)$

$(ii) \Rightarrow e^x \neq 0 \Rightarrow \sin y = 0 \Rightarrow y = \sin^{-1}(0) \Rightarrow y = n\pi ; n = 0, \pm 1, \pm 2, \dots$

$(i) \Rightarrow e^x \cos y = -1 \Rightarrow e^x \cos n\pi = -1 \Rightarrow e^x (-1)^n = -e^0$

$\Rightarrow (-1)^n = -1$ if $n = (2k + 1)$; $k = 0, \pm 1, \pm 2, \pm 3, \dots$

Thus $e^x = e^0 \Rightarrow x = 0$ and therefore $y = n\pi = (2k + 1)\pi$

Hence $z = x + iy = (2k + 1)\pi i$

For Sufficient Condition:

Given $z = (2k + 1)\pi i$ then we have to show that $e^z = -1$

Let $e^z = e^{(2k+1)\pi i} = \cos(2k + 1)\pi + i \sin(2k + 1)\pi = -1 + 0i$

$\Rightarrow e^z = -1$ as required

Property: Prove that $e^z = i$ iff $z = \left(\frac{1}{2} + 2n\right)\pi i$; $n = 0, \pm 1, \pm 2, \dots$

Solution: Consider $e^z = i$ Then we have to prove $z = \left(\frac{1}{2} + 2n\right)\pi i$

Now as $e^z = i \Rightarrow e^x e^{iy} = i \Rightarrow e^x (\cos y + i \sin y) = i$

$\Rightarrow e^x \cos y + i e^x \sin y = 0 + i$

$e^x \cos y = 0 \dots\dots(i) \quad e^x \sin y = 1 \dots\dots(ii)$

$(i) \Rightarrow e^x \neq 0 \Rightarrow \cos y = 0 \Rightarrow y = \cos^{-1}(0) \Rightarrow y = \left(\frac{1}{2} + 2n\right)\pi$

$(ii) \Rightarrow e^x \sin y = 1 \Rightarrow e^x \sin \left(\frac{1}{2} + 2n\right)\pi = 1 \Rightarrow e^x (1)^n = e^0$

Thus $e^x = e^0 \Rightarrow x = 0$

Hence $z = x + iy = 0 + \left(\frac{1}{2} + 2n\right)\pi i$

For Sufficient Condition:

Given $z = \left(\frac{1}{2} + 2n\right)\pi i$ then we have to show that $e^z = i$

Let $e^z = e^{\left(\frac{1}{2} + 2n\right)\pi i} = \cos \left(\frac{1}{2} + 2n\right)\pi + i \sin \left(\frac{1}{2} + 2n\right)\pi = 0 + i$

$\Rightarrow e^z = i$ as required

PERIODIC FUNCTION: A function $f(z)$ is said to be periodic if there exists a non – zero constant λ (complex) such that $f(z + \lambda) = f(z)$

Thus λ is called period of the function.

Property: Prove that e^z is periodic function.

Solution: if $f(z) = e^z$ then we have to show the periodic value of e^z

Consider $f(z + \lambda) = f(z) \Rightarrow e^{z+\lambda} = e^z$

If $z = 0$ then $\Rightarrow e^{0+\lambda} = e^0 \Rightarrow e^\lambda = 1$

Since λ is complex valued therefor $\lambda = \alpha + i\beta \dots\dots\dots(i)$

Then $e^\lambda = 1 \Rightarrow e^{\alpha+i\beta} = 1 \Rightarrow e^\alpha (\cos \beta + i \sin \beta) = 1 + 0i$

$e^\alpha \cos \beta = 1 \dots\dots(ii) \quad e^\alpha \sin \beta = 0 \dots\dots(iii)$

$(iii) \Rightarrow e^\alpha \neq 0 \Rightarrow \sin \beta = 0 \Rightarrow \beta = \sin^{-1}(0) \Rightarrow \beta = n\pi ; n = 0, \pm 1, \pm 2, \dots$

$(ii) \Rightarrow e^\alpha \cos \beta = 1 \Rightarrow e^\alpha \cos n\pi = 1 \Rightarrow e^\alpha (-1)^n = e^0$

$\Rightarrow (-1)^n = 1$ if $n = 2k ; k = 0, \pm 1, \pm 2, \pm 3, \dots\dots\dots$

Thus $e^\alpha = e^0 \Rightarrow \alpha = 0$ and therefore $\beta = n\pi = 2k\pi$

Hence $\lambda = \alpha + i\beta = 2k\pi i$

Now e^z is periodic functions as

$f(z) = f(z + \lambda) \Rightarrow e^z = e^{z+\lambda} = e^{z+2k\pi i} = e^z (\cos 2k\pi + i \sin 2k\pi)$

$\Rightarrow e^{z+\lambda} = e^z (1 + 0i) \Rightarrow e^{z+\lambda} = e^z$ Thus $\lambda = 2k\pi i$ is period of e^z

For $k = 1$, $\lambda = 2\pi i$ is Simple, Fundamental or Primitive period of e^z and the other periods are $\lambda = 4\pi i, \lambda = 6\pi i, \lambda = 8\pi i$ etc

Remark: The periodicity of the exponential function does not appear in the real domain, since the periods of the function are all imaginary.